

Capacity estimation of lithium-ion batteries based on optimized charging voltage section and virtual sample generation

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HIGHLIGHTS

- Seven health factors are extracted from data domain and model domain.
- Quantum particle swarm optimization is used to extract optimal voltage section.
- Multi-distribution mega-trend-diffusion is used to generate virtual samples.
- Long short-term memory neural network is applied for capacity estimation.
- The problems of non-fully charging and insufficient samples can be overcome.

ARTICLE INFO

Keywords:

Lithium-ion batteries
Capacity estimation
Optimal voltage section
Virtual sample generation

ABSTRACT

Accurate capacity estimation of a lithium-ion battery is crucial to predict its state of health and remaining useful life, determining the reliability and safety of the battery system directly. However, existing data-driven methods usually are based on full battery charging or discharging curves, which are hard to meet in practice because of range anxieties and arbitrary charging behaviors of users. Meanwhile, the lack of sufficient datasets for model training limits the performance of a data-driven model. In this paper, a capacity estimation method for lithium-ion batteries based on an optimized charging voltage section and virtual sample generation is proposed. In the method, characteristics of full and sectional capacity degradation are analyzed to evaluate the feasibility of capacity estimation using a section of charging voltage curve. The extraction of health indicator and optimization of battery voltage range are introduced. A multi-distribution global trend diffusion algorithm is used to generate virtual samples. Based on a long short-term memory neural network, battery capacity estimation model is constructed. Experimental verifications on NASA, CALCE, and Oxford battery datasets are carried out. Results show that the proposed method not only reduces the length of required voltage section for capacity estimation, but also improves the estimation accuracy in comparison to utilization of the full charging curve. Under the operating conditions of NASA B6, CALCE CS36 and Oxford C3, the root mean square error (RMSE) of battery capacity estimation is lower than 1.48%.

1. Introduction

Lithium-ion batteries have the advantages of high energy and power densities, low self-discharge rate, long lifespan, and high environmental adaptability. They have been widely used in many fields, such as consumer electronics, electric vehicles, and smart grids [1]. During usage, lithium-ion batteries inevitably suffer from performance degradation, such as reduction in available capacity and increase in internal

resistance. In particular, if the performance degrades to a certain stage, the battery internal resistance will increase greatly, increasing the risk of thermal runaway. In addition, some key states of the lithium-ion battery, such as state of charge (SOC), state of health (SOH) [2], and remaining useful life (RUL) [3] are highly related to the available capacity. Therefore, accurate capacity estimation is very important for the operating safety and reliability of the lithium-ion battery.

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<https://doi.org/10.1016/j.apenergy.2022.120516>

Received 23 September 2022; Received in revised form 12 November 2022; Accepted 9 December 2022

Available online 18 December 2022

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Nomenclature			
r	correlation coefficient	$Skew_L$	left skewness
$Var1$	variable 1	$Skew_U$	right skewness
$Var2$	variable 2	N_L	number of samples lower than CL
I_c	charging current	N_U	number of samples greater than CL
R_o	ohmic resistance	m	correction factor
R_p	polarization resistance	$\hat{S}_x^2$	variance
C_p	polarization capacitance	LB	lower bound
U_t	terminal voltage	UB	upper bound
U_{oc}	open-circuit voltage	min	minimum in X
F	sum of correlations	max	maximum in X
V_a	left point of the voltage section	$g(\cdot)$	fitness function
V_b	right point of the voltage section	$fitness$	similarity between virtual and raw sample
v_i	the flying direction of the i -th particle	z	original sample
w	inertia factor	z_i	the i -th virtual sample
$rand_1$	a random in the (0, 1)	f	Forget gate
$rand_2$	a random in the (0, 1)	i	input gate
p_i	the current position of the i -th particle	O	output gate
P_i	update of the i -th particle position	C	cell state
$pbest$	historical best position	h	estimation result
$gbest$	the group best position	HF_0	seven HF's in the origin dataset
L_1	local learning rate	HF_N	seven HF's in the N -th virtual dataset
L_2	global learning rate	HF_n	seven HF's in n -th cycle
$mbest$	average history best position	$capacity_n$	capacity in n -th cycle
φ	a random number in (0,1)	q	real capacity
u	a random number in (0,1)	$\hat{q}_i$	estimated capacity
α	compression expansion coefficient	$\bar{q}$	average value of the real capacity
X	original sample set	RMSE	root mean square error
CL	data center	MAPE	absolute percentage error
x_i	the i -th data in X	R^2	goodness of fit

1.1. Literature review

Currently, there are mainly-two types of methods for online battery capacity estimation, namely the model-driven methods and the data-driven methods. The model-driven methods can be further divided into two categories: the electrochemical model-based method [4] and the equivalent circuit model-based method [5,6]. The electrochemical model can simulate the internal working mechanism of the lithium-ion battery accurately. Nevertheless, it is limited by complex structure and low calculation efficiency. In contrast, the equivalent circuit model has a simple structure and high calculation efficiency, but its accuracy and information richness are not as good as the electrochemical model. However, it is hard to establish an accurate and commonly used electrochemical/equivalent circuit model for different battery types and different operating conditions, which highly limits the applications of the model-driven methods.

The data-driven method can learn a capacity estimation model directly from the operating data of the lithium-ion battery. It has been widely concerned for battery capacity estimation and is highly potential in practical applications [7]. The data-driven method regards the battery system as a "black box", and needs not to know the complex electrochemical reactions inside the battery. It is very crucial for the data-driven method to extract appropriate health factors (HFs) that can well characterize the law of battery capacity degradation from the voltage, current and temperature, and to construct a high-precision machine learning model between the extracted HFs and the capacity. The HF extraction is a key foundation of battery capacity estimation, which significantly affects the capacity estimation result. At present, the commonly used HF extraction methods include incremental capacity analysis (ICA) [8,9], differential voltage analysis (DVA) [10,11], and differential thermal voltammetry (DTV) [12,13].

The machine learning model also plays an important role in the accurate estimation of battery capacity. With the rapid development of machine learning and artificial intelligence, more and more algorithms, such as support vector machine (SVM) [14,15], correlation vector machine [16], Gaussian process regression [17], and neural networks [18,19], are introduced to battery capacity estimation. Recently, some neural network models that are suitable for time series analysis have been successfully used to estimate the capacity or SOH of lithium-ion batteries. Tan et al. [20] proposed a SOH estimation model based on a long short-term memory (LSTM) network and introduced transfer learning to enhance the model generalization for various working conditions and battery batches. Li et al. [21] constructed a deep LSTM network model for lithium-ion battery capacity estimation and verified its ability to process input data sequences with different lengths and noises.

The aforementioned existing studies are based on the full charging/discharging data of lithium-ion batteries. However, the batteries in real-world conditions, especially for electric vehicle applications, are neither fully discharged nor fully charged. In most cases, the battery charging starts when the SOC is about 5 %-20 % because of range anxiety and stops as the SOC reaches 80 %-100 % to avoid over-charging. As a result, estimating the available capacity based on a charging/discharging voltage section has practical significance. Park et al. [22] selected the charging capacity in a fixed SOC interval at the constant current stage as the HF and realized capacity estimation using a multi-layer perception. Yang et al. [23] adopted the charging capacity in the voltage section of 3.95 V-4.15 V as the HF, and estimated the battery SOH by the least square method. These studies realized battery capacity estimation based on sectional charging data. However, they failed to discuss the selection of the optimal section, resulting in limited estimation accuracy. Zheng et al. [24] utilized a particle swarm optimization algorithm to search the

optimal section of charging curve for HF extraction and established a homogeneous linear model between the HF and the capacity. Meng et al. [25] employed a non-dominated sorting genetic algorithm II (NSGA-II) to select the optimal multi-voltage sections, in which both the section length and estimation error were taken into consideration. Although these two studies considered the selection of the optimal voltage section, they adopted a linear model to capture the relationship between the sectional capacity and the full capacity, which may lead to large estimation errors, especially for different battery cells under different operating conditions. In addition, they only selected the sectional capacity as a single HF, which cannot fully reflect the capacity degradation feature of the lithium-ion battery and is highly dependent on the data, thus resulting in poor generalization ability of the model.

Furthermore, because of the long cycling life, various operating conditions and inevitable cell inconsistency of lithium-ion batteries, it is hard to acquire sufficient test data for training a precise and generalized estimation model. To this end, virtual sample generation technologies have been introduced to extend the small sample dataset. Kang et al. [26] proposed an overall trend virtual sample generation (OT-DEVSG) algorithm for generating a series of similar virtual battery datasets based on a single battery dataset. Zhang et al. [27] designed a multi-population differential evolution (MP-DE) method to expand the raw battery data, and investigated the selection of qualified virtual battery data. However, the above studies are based on the full charging curve of lithium-ion batteries. In addition, they established the estimation model based on characteristics of the capacity degradation curve, lacking well-designed HF. As a result, they would have limited flexibility and generalizability in practical applications.

1.2. Motivation and main contributions

This paper proposes a capacity estimation method for lithium-ion batteries based on an optimized charging voltage section and a virtual sample generation algorithm. The motivation is to achieve reliable capacity estimation of lithium-ion batteries based on limited test data and regardless of unexpected changes in discharging conditions. The innovations and contributions of the paper mainly include:

The HF is extracted from both data domain and model domain. The HF in data domain is extracted based on the ICA curve and charging voltage section, while the HF in model domain is extracted from the first-order equivalent circuit model. By fusing the data-domain and model-domain HF, the proposed method can reduce the dependency on data and improve the robustness to HF abnormalities.

(2) The charging voltage section is optimized using a QPSO algorithm. A nonlinear programming model for voltage section optimization is constructed. The optimization objective is to maximize the sum of absolute Pearson coefficients between each single HF and the battery capacity. Also, constraints are added on both SOC and the length of voltage section in consideration of users' charging behaviors and safety concerns. In this way, the proposed method can achieve high-precision capacity estimation based on a very small section of charging voltage curve only and well meet the requirements in practical applications.

(3) A multi-distribution global trend diffusion (MD-MTD) algorithm is developed to augment the datasets for model training. The generated data are divided into extended area and direct observation area. To generate virtual sample points, a uniform distribution is adopted in the extended area, while a triangular distribution is used in the direct observation area. Also, a correction factor is added to avoid unreasonable expansion in the extended area, making up for drawbacks of the conventional MTD algorithm. It is benefit for improving the generalization and accuracy of the neural network model for battery capacity estimation.

(4) The proposed method is evaluated on three battery datasets

comprehensively. Comparison studies, including 1) capacity estimation with the full charging curve, a non-optimal voltage section, and the optimized voltage section, 2) capacity estimation with and without virtual samples are implemented. Results demonstrate the effectiveness of the proposed method in reduction of the voltage section length and improvement of capacity estimation accuracy compared with the full charging curve.

1.3. Paper organization

The remainder of the paper is organized as follows. Section 2 introduces the battery datasets used in this paper, and analyzes their capacity degradation characteristics at both full and partial charging levels. Section 3 illustrates the principle of the proposed method in detail. Experimental verification is carried out and discussed in Section 4. Finally, main conclusions of the paper are summarized in Section 5.

2. Battery datasets

Three public datasets including NASA, CALCE, and Oxford are employed to evaluate the performance of the proposed method. These three public datasets are popular and all from the leading battery laboratories, the experiment and data record are standard, which is conducive to the comparison between different methods.

This section introduces experimental processes of the three datasets, and analyzes the feasibility of capacity estimation based on partial charging voltage curve.

2.1. Test procedure and capacity degradation

(1) NASA lithium-ion battery dataset.

NASA lithium-ion battery dataset is provided by NASA Ames Research Center. In this paper, three batteries, B5 (training set), B6 (test set), and B7 (test set) are selected. Their rated capacity is 2 Ah, and the cathode material is lithium cobalt oxide (LiCoO₂). The battery life aging test is conducted at room temperature. The experimental process is as follows: the batteries are charged with a constant current of 1.5 A until their voltage reaches the upper cut-off value of 4.2 V, followed by a constant voltage stage till the current is below 20 mA, then the three batteries are discharged with a constant current of 2 A to a voltage of 2.7 V, 2.5 V and 2.2 V, respectively.

The capacity degradation curves of B5, B6, and B7 are shown in Fig. 1 (a), and the changes of B5 in voltage and current with the increase of life cycles are shown in Fig. 1 (b) and (c), respectively. The changes of the other two batteries in voltage and current are similar to B5.

(2) CALCE lithium-ion battery dataset.

The CALCE lithium-ion battery dataset comes from the center for Advanced Life Cycle Engineering (CALCE) of the University of Maryland. In this paper, three batteries, CS35 (training set), CS36 (test set), and CS37 (test set) are selected. Their rated capacity is 1.1 Ah, and the cathode material is lithium cobalt oxide (LiCoO₂). The battery life aging test is conducted at room temperature. The experimental process is as follows: the batteries are charged with a constant current of 0.55 A until the voltage increases to the upper limitation of 4.2 V, followed by a constant voltage charging stage until the current less than 20 mA; then the batteries are discharged with a constant current of 2 A until the voltage decreases to 2.7 V.

The capacity degradation curves of batteries CS35, CS36, and CS37 are shown in Fig. 2 (a), and the changes of CS35 in voltage and current with the increase of cycle are shown in Fig. 2 (b) and (c), respectively. The changes of the other two batteries in voltage and current are similar to CS35.

(3) Oxford lithium-ion battery dataset.

Oxford lithium-ion battery dataset comes from the battery Intelligence Laboratory of Oxford University. Three batteries, C1 (training set), C3 (test set), and C8 (test set) are selected in this study. Their rated

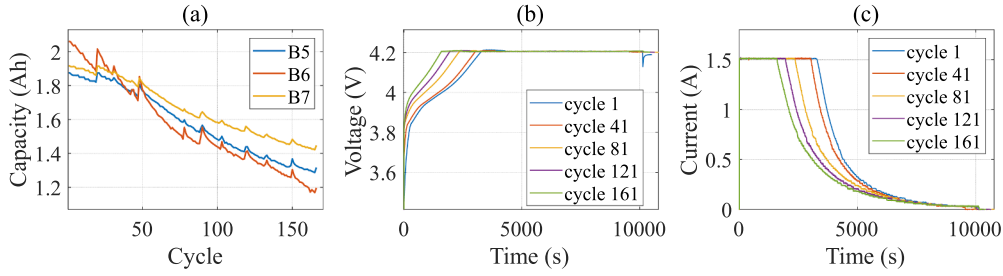


Fig. 1. Details of the NASA dataset: (a) capacity degradation curves of B5, B6 and B7 batteries; (b) voltage curve of B5; (c) current curve of B5.

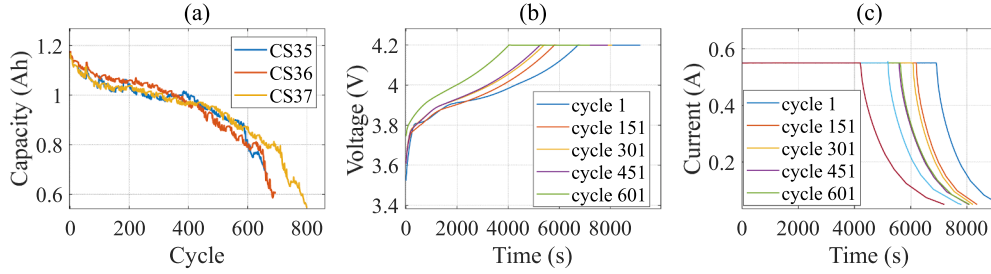


Fig. 2. Details of the CALCE dataset: (a) capacity degradation curve of CS35, CS36 and CS37; (b) voltage curve of CS35; (c) current curve of CS35.

capacity is 0.74 Ah, and the cathode material is lithium cobalt oxide (LiCoO₂). The batteries are tested at 40 °C and the test process is as follows: the batteries are firstly charged with a constant current of 0.74 A to their upper cut-off voltage of 4.2 V, and then they are discharged with a constant current of 2 A to a lower cut-off voltage of 2.7 V.

The capacity degradation curves of batteries C1, C3, and C8 are plotted in Fig. 3 (a), and the changes of C1 in voltage and current with the increase of cycle are plotted in Fig. 3 (b) and (c), respectively. The changes of the other two batteries in voltage and current are similar to C1.

2.2. Degradation characteristics of sectional charging capacity

To demonstrate the feasibility of capacity estimation based on a charging voltage section, this section selects two voltage sections, 3.8 V–4.0 V and 3.9 V–4.1 V as two examples for case studies. Capacity degradation curves corresponding to these two voltage sections are plotted in Fig. 4. It is evident that the sectional capacity degradation curve exhibits a very similar trend with the full capacity degradation curve, and it can capture the capacity regeneration feature of the battery. To quantify the correlations between the sectional capacity and the full capacity, the Pearson coefficient is employed herein, and it is formulated as:

$$r = \frac{\sum_{i=1}^n (Var1_i - \overline{Var1})(Var2_i - \overline{Var2})}{\sqrt{\sum_{i=1}^n (Var1_i - \overline{Var1})^2} \sqrt{\sum_{i=1}^n (Var2_i - \overline{Var2})^2}} \quad (1).$$

where *Var1* and *Var2* denote two variables, $\overline{Var1}$ and $\overline{Var2}$ are the

average of *Var1* and *Var2*, respectively.

Pearson coefficients between the sectional capacity and the full capacity are detailed in Table 1. It can be seen that correlations between the sectional capacity and the full capacity are always greater than 0.96. Therefore, it is feasible to estimate the full battery capacity based on a section of voltage curve. Also, it is clear that for all of the three battery datasets, correlation coefficients in the section of 3.8 V–4.0 V are always higher than those in the section of 3.9 V–4.1 V. In other words, different voltage sections perform differently in capturing the law of the full capacity degradation. Accordingly, it is crucial to find an optimal voltage section for accurate capacity estimation, which will be detailed in Section 3.

As can be seen from Table 1, the correlations between sectional and full capacity are very high, about 0.99, which demonstrate that sectional capacity can be adopted as a HF to characterize the law of capacity degradation. However, a single HF is easy to cause over fitting, that is, which is only good for training set, but poor for test set, leading to poor generalization ability of the model. To ensure the generalization ability of the model, it is necessary to extract the HFs from multi-dimension to maintain appropriate HF redundancy.

3. Proposed method

In practical applications, charging the battery is usually full of randomness and uncertainty. For example, users would start charging the battery when its SOC is about 10 %–20 %, instead of waiting for the

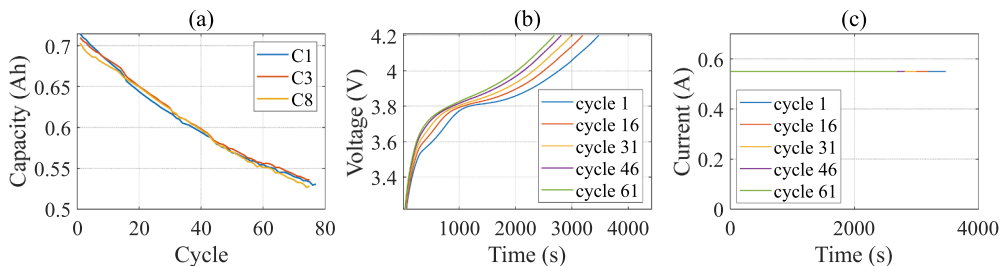


Fig. 3. Details of the Oxford dataset: (a) capacity degradation curves of C1, C3 and C8; (b) voltage curve of C1; (c) current curve C1.

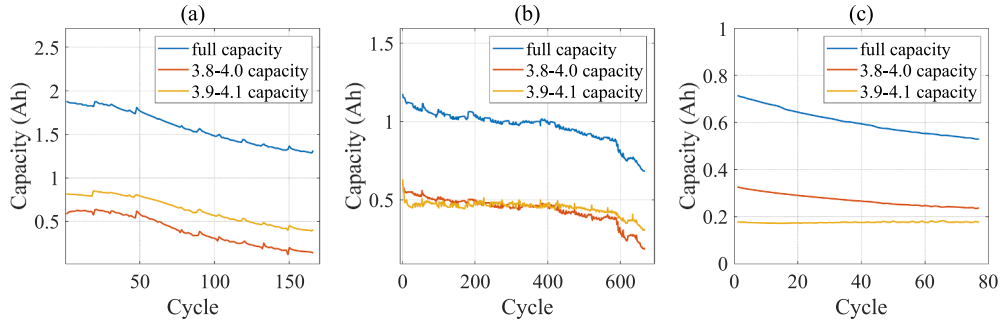


Fig. 4. The full capacity and sectional capacity: (a) NASA; (b) CALCE; (c) Oxford.

Table 1

Pearson coefficients between sectional capacity and full capacity.

Battery	3.8 V-4.0 V	3.9 V-4.1 V
NASA B5	0.9967	0.9928
CALCE CS35	0.9971	0.9647
Oxford C1	0.9999	0.9973

full discharge of the battery (SOC = 0) because of range anxieties, and stop charging before the SOC reaches 100 % for the security consideration of over-charging. In particular, at a fast-charging scenario, the battery would be charged to 80 % SOC only or even a lower SOC level. In summary, the full charging curve of a battery cannot always be available. Therefore, this paper proposes a capacity method for lithium-ion batteries based on a charging voltage section. The flowchart of the proposed method is illustrated in Fig. 5. Overall, it can be divided into two stages: offline training and online estimation. The former includes

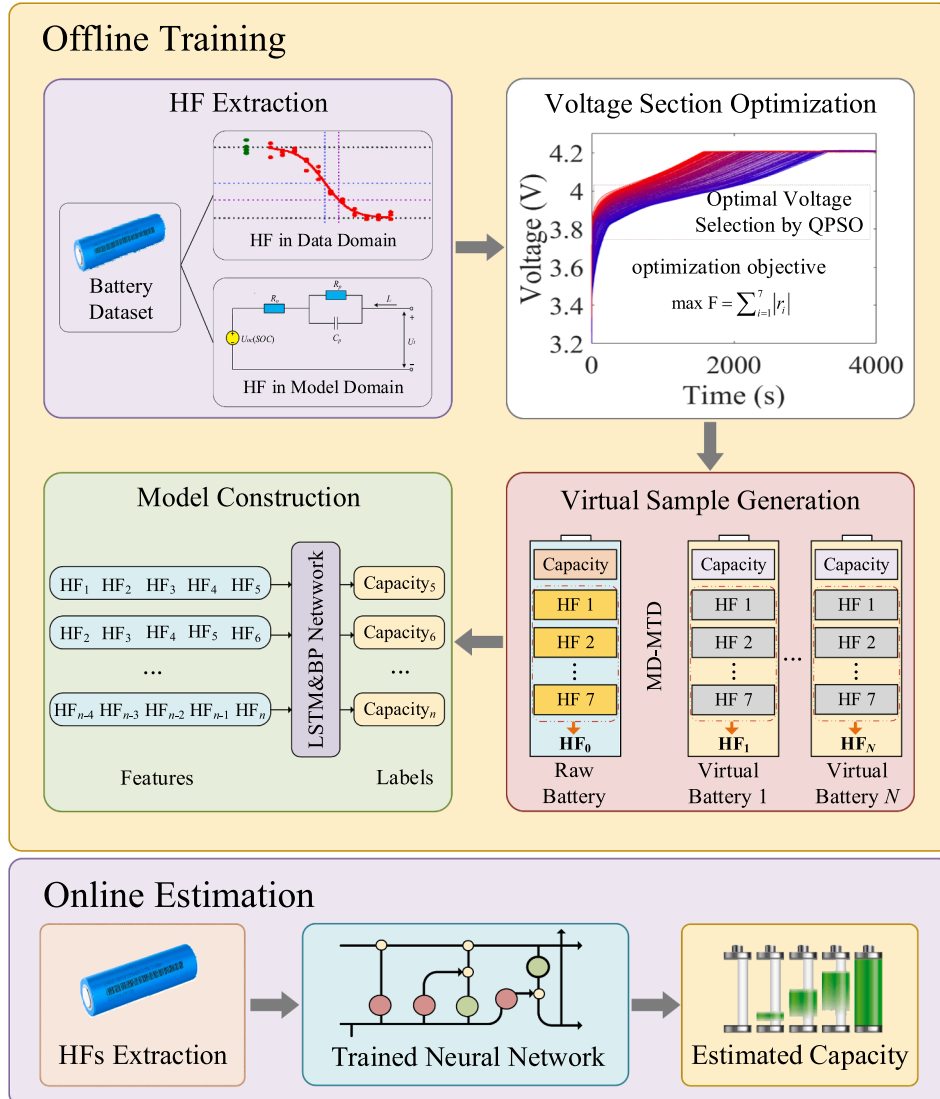


Fig. 5. The flowchart of the proposed method.

four steps, including HFs extraction, which are the foundation of high accuracy estimation; voltage section optimization, while ensuring HF quality and user's charging coverage possibilities; virtual samples generation, extending the original sample to enhance the generalization ability of the model; capacity estimation model construction, a LSTM&BP neural network is utilized to construct a high accuracy estimation model.

3.1. HF extraction

During the charging and discharging process of the battery, Li^+ are repeatedly detaching between the positive and negative electrodes, which is accompanied by side-reactions, resulting in irreversible loss of the chemical substances in the battery. With the increasing of cycles, the side-reactions gradually intensify. In the micro view, the solid electrolyte interphase (SEI) will grow, and the electrodes will dissolve. In the macro view, the internal resistance will increase, and the capacity will decrease [28].

Because the internal resistance and capacity cannot be acquired directly, it is very crucial for a data-driven method to extract appropriate HFs that can well characterize the law of capacity degradation. Usually, HFs are directly and only extracted from charging/discharging data, such as voltage, current, and temperature. As a result, they have strong dependence on the data. In practical applications, however, the charging/discharging data inevitably suffer from measurement noises, incompleteness, inconsistency due to varied working conditions, leading to the instability of HFs. To reduce the dependence of HFs on data, this paper extracts HFs from both the data domain and the model domain.

The HFs in data domain are extracted from the incremental capacity analysis (ICA) curve and the time-voltage curve in charging process, which are illustrated in Figs. 6 and 7, respectively. It is clear from Fig. 6 that for a same voltage range, the charging time and charging capacity reduce, while the voltage rise rate (curve slope) increases with the increase of cycle number. The ICA can transform the gentle rising process in the voltage curve into an obvious rising process, which is conducive to finding more workable HFs. As can be seen from Fig. 7, the peak value of the ICA curve decreases and shifts to the right with the increase of cycle number.

In summary, these four indexes, namely, charging capacity, voltage rise rate, peak value of ICA curve, and peak position of ICA curve are closely related to the battery aging, so they are chosen to be HFs in data domain.

Although the HFs in data domain can well indicate the law of battery capacity degradation, it is benefit for improving the richness and

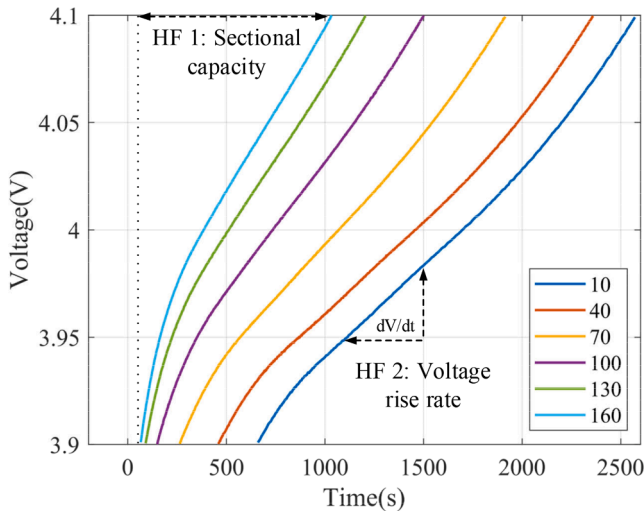


Fig. 6. Examples of the time-voltage curve.

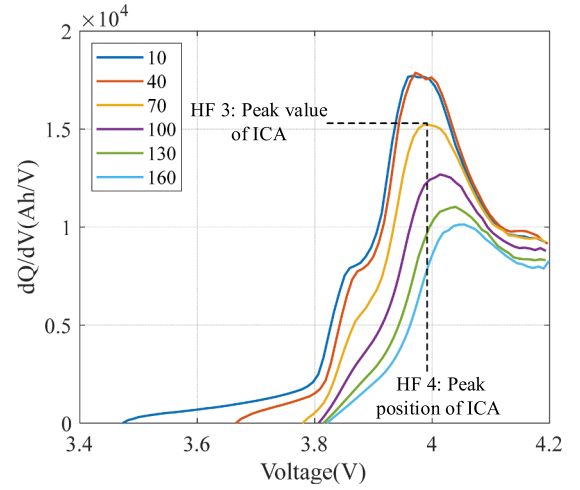


Fig. 7. Examples of the ICA curve.

reliability of HFs to add HFs in model domain considering the fluctuation and loss of data. The introduction of HFs in model domain to a data-driven model can improve its robustness and generalization. To obtain HFs in model domain, the commonly used first-order resistor-capacitor (RC) equivalent circuit model is applied in this paper, and its parameters are identified by the least square algorithm [29].

The schematic of the first-order RC is illustrated in Fig. 8, where I_c represents charging current, R_o denotes ohmic resistance, R_p and C_p are polarization resistance and capacitance, respectively, U_t is terminal voltage, and U_{oc} denotes open-circuit voltage, which varies with the state of charge (SOC) of the battery. R_o , R_p and C_p are adopted to be HFs in model domain because they are highly sensitive to battery aging [30]. The extracted seven HFs are summarized in Table 2.

Although the above seven HFs inevitably have redundancy, it is necessary. In the practical application, multi-HF are utilized to estimate capacity simultaneously, which can reduce the impact of a certain HF failure on the estimation results, and increase the robustness of the method.

3.2. Voltage section optimization

As shown in Table 1, different voltage sections perform differently in capturing the law of the full capacity degradation. In addition, some HFs are highly dependent on the voltage range. Taking the ICA curve as an example, the peak value locates around 4 V. Therefore, it is crucial to choose an optimized voltage section for enhancing the correlation between HFs and capacity.

The optimization objective is set to maximize the sum of the absolute Pearson coefficients among the seven HFs and the capacity, which is formulated as:

$$\max F = \sum_{i=1}^7 |r_i|(2).$$

where r_i is the correlation coefficient between the i -th HF and the

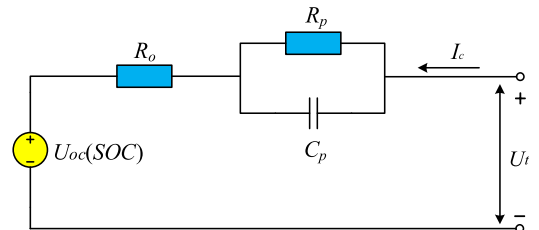


Fig. 8. The schematic of the first-order RC battery equivalent circuit model.

Table 2

Extracted HF's in both data-domain and model-domain.

Domain	Label	Name	Explanation
Data	HF 1	Sectional capacity	Charging capacity in the selected voltage section
	HF 2	Voltage rise rate	The ratio of the voltage rise to charging time
	HF 3	Peak value of ICA curve	Maxima of the ICA curve in the voltage section
	HF 4	Peak position of ICA curve	Voltage referring to the maxima of the ICA curve
Model	HF 5	Ohmic resistance	Combination of electrode material, electrolyte, diaphragm resistance, and contact resistance
	HF 6	Polarization resistance	Resistance caused by polarization in the electrochemical reaction
	HF 7	Polarization capacitance	Capacitance caused by polarization in the electrochemical reaction

capacity.

In practical applications, electric vehicle users are very likely to start charging the battery before SOC = 0 % because of range anxieties, and stop charging before SOC = 100 % to avoid over-charging, especially under a fast-charging condition. Also, the selected voltage range should be as small as possible in order to reduce the computation cost as well as to improve the availability of the required data for capacity estimation. In consideration of the above factors, this paper adds constraints on both the SOC and the length of voltage section as follows:

$$\begin{cases} 10\% \leq \text{SOC} \leq 80\% \\ 0.05V \leq V_b - V_a \leq 0.25V \end{cases} \quad (3).$$

where V_a and V_b are the left and right end points of the voltage section, respectively.

According to Eqs. (2) and (3), the voltage section optimization is a multi-constraint nonlinear programming problem, which will be resolved by a QPSO algorithm in this paper.

Particle swarm optimization (PSO) is an evolutionary computation algorithm originally proposed by J. Kennedy and R. C. Eberhart in 1995 [31]. The basic principle of the PSO algorithm is as follows: 1) Assuming that there are a group of particles flying at a certain speed in a search space, every single particle is considered to be a possible solution of the problem to be solved, and the speed of each particle is expressed by a vector with size and direction; 2) At each iteration, the particles employ both their own and group's flight experience to adjust their states, and to calculate their individual fitness; 3) Then the individual extremum and group extremum can be obtained by comparing all the individual fitness. The status update of i -th particle is formulated as follows:

$$v_i = w \times v_i + L_1 \times \text{rand}_1 \times (pbest_i - p_i) + L_2 \times \text{rand}_2 \times (gbest_i - p_i) \quad (4).$$

where v_i represents the flying direction of the i -th particle, $pbest_i$ denotes the historical best position of the i -th particle, $gbest_i$ is the group best position at i -th step, p_i represents the current position of the i -th particle, rand_1 and rand_2 are two random numbers in the range of (0, 1), w denotes the inertia factor, L_1 and L_2 are local and global learning rates, respectively.

However, there are mainly-two limitations in a classical PSO algorithm. First, the particle must be in bound state to avoid explosion and guarantee convergence, which possibly makes the PSO algorithm fall into a local optimization solution. Second, it is trial to control and select the five parameters, including w , rand_1 , rand_2 , L_1 and L_2 . To this end, sun et al. proposed a QPSO algorithm in 2004 [32]. From the perspective of quantum theory, the QPSO algorithm regards each particle as a quantum particle. The state of each quantum particle is no longer expressed by a position vector and a direction vector, but depicted by a wave function. The probability of a particle appearing at a certain position is expressed by the probability density function, not limited to the established orbit function. The particle update steps of the QPSO algorithm are as follows:

1) Calculate the average particle history best position.

$$mbest = \frac{1}{n} \sum_{i=1}^n pbest_i \quad (5).$$

where n represents the number of particles.

2) Particle position update.

$$p_i = P_i \pm \alpha |mbest - p_i| \times \ln\left(\frac{1}{u}\right) \quad (6).$$

$$P_i = \varphi \cdot pbest_i + (1 - \varphi) \cdot gbest$$

where φ and u are two random numbers in (0,1), α is a compression expansion coefficient, which is generally smaller than 1. In this paper, α is set to 0.8.

3.3. Virtual sample generation

The battery capacity degradation test is time-consuming and it is impossible to cover all the conditions in practical applications. As a result, the available datasets for model training usually are limited and incomplete, leading to weak generalization ability of the neural network model. To this end, this paper introduces a multi-distribution global trend diffusion (MD-MTD) method to expand the upper and lower bounds of raw data so as to increase the data samples for model training.

MD-MTD is a kind of virtual sample generation technology. When the number of sample is insufficient, the generalization ability, robustness, and accuracy of machine learning will be greatly affected. Efforts have been made to scientifically and reasonably expand the scale of samples according to the raw data and enhance the performance of machine learning models [33,34]. Li [35] proposed the global trend diffusion (MTD) technology in 2007. MTD is a virtual sample generation technology based on global fuzziness and information diffusion, which estimates the distribution population through a triangular membership function. However, the triangle distribution used in MTD is too simple to describe the deep-seated characteristics and complex overall distribution of raw data samples. Later, Zhu et al. [36] proposed a multi-distribution overall trend diffusion technology (MD-MTD) in 2016. In the extending area, a uniform distribution is employed, while in the direct observation area, a triangular distribution is used to generate virtual sample points as shown in Fig. 9. Furthermore, a correction factor m is added to avoid unreasonable expansion of the expanded area, which makes up for limitations of the classical MTD.

The basic principle of MD-MTD algorithm is summarized as follows:

Step 1: Calculate the data center, left skewness, right skewness and variance of the original sample set $X = \{x_1, x_2, \dots, x_n\}$, where n denotes the total amount of data in X .

(1) Data center, CL :

$$CL = \begin{cases} x_{(n+1)/2}, & n = 1, 3, 5, \dots \\ \frac{1}{2}(x_{n/2} + x_{(n+1)/2}), & n = 2, 4, 6, \dots \end{cases} \quad (7).$$

(2) Left skewness, $Skew_L$:

$$Skew_L = \frac{N_L}{N_L + N_U + m} \quad (8).$$

where N_L and N_U represent the number of samples lower and greater than CL , respectively; m is a correction factor, which usually is set to be $n/10$.

(3) Right skewness, $Skew_U$:

$$Skew_U = \frac{N_U}{N_L + N_U + m} \quad (9).$$

(4) Variance, $\hat{S}_x^2$:

$$\hat{S}_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \quad (10).$$

Step 2: Calculate the acceptable boundary of the original sample set. The lower and upper bounds of the acceptable range of X can be separately formulated as:

$$LB = \begin{cases} CL - Skew_L \times \sqrt{-2 \frac{\hat{S}_x^2}{N_L} \times \ln(10^{-20})}, & LB \leq \min \\ \min, & LB > \min \end{cases} \quad (11).$$

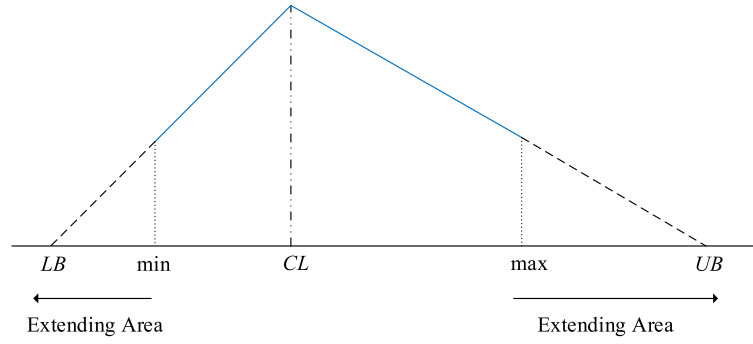


Fig. 9. Schematic of the MD-MTD data distribution.

$$UB = \begin{cases} CL + Skew_U \times \sqrt{-2 \frac{\hat{S}_x^2}{N_U} \times \ln(10^{-20})}, & UB \geq \max \\ \max, & UB < \max \end{cases} \quad (12).$$

Step 3: A triangular distribution is adopted to describe the data distribution in the direct observation area $[\min, \max]$ of the sample set X , and a uniform distribution is used to generate virtual sample points in the extending areas, $[LB, \min]$ and $[\max, UB]$.

In this paper, the MD-MTD algorithm is firstly utilized to determine the acceptable range of the raw samples, and then the QPSO algorithm introduced in section 3.2 is employed to select the virtual samples that meet the constraints as follows:

$$\begin{cases} LB \leq z_i \leq UB \\ g(z) = \sum_{i=1}^n \frac{(z - z_i)}{n} \\ g(z) \leq fitness \end{cases} \quad (13).$$

where z_i represents the i -th virtual sample, z represents the original sample, $g(z)$ is the fitness function, and $fitness$ is a pre-determined value to control the similarity between the virtual sample and the raw sample.

To determine the value of $fitness$, 5 candidate values are utilized to generate the virtual samples, respectively, shown in Fig. 10. The $fitness$ value controls the similarity between the virtual samples and the original sample. When the $fitness$ is small, the original sample and the virtual samples are very similar, which will lead to the virtual samples hard to work. With the rise of $fitness$ value, the difference increases, and the generalization function of virtual samples gradually appears. However, when the $fitness$ value is too large, the difference between the two is huge, and even the trend is no longer same. Virtual samples have been harmful to the model establishment. Therefore, this paper set the $fitness$ as 0.01.

Taking the capacity degradation curve as an example, the generated virtual capacity curves based on the original capacity are illustrated in Fig. 11. Similarly, the other virtual HFs can be generated in this way. It can be seen from Fig. 11 that the virtual capacity curves exhibit approximate trend with the original capacity curve, but have different values, thus augmenting the datasets very well.

Then HFs (input) and the capacity (output) are combined as virtual battery samples. In such a way, N sets of virtual batteries can be

generated using a set of battery characteristics, as shown in Fig. 12, where HF_0 denotes the vector composed of the seven HFs extracted from the original battery dataset corresponding to the optimal voltage section, and HF_N represents the vector of HFs extracted from N -th generated virtual battery dataset. To make a tradeoff between the computation efficiency and data richness, N is set to 50 in this paper.

3.4. Capacity estimation model

To establish a data-driven model between HFs and the capacity, the LSTM and BP neural networks are adopted in this paper. LSTM network is a recurrent neural network proposed by Hochreiter and Schmidhuber [37] in 1997, which makes up for the traditional RNN network's inability to deal with long-term dependence and avoids gradient explosion and gradient disappearance. It has been widely used in time series prediction and natural language processing. In general, the LSTM network introduces the input gate, output gate, and forget gate into the cell for updating, saving and forgetting information, respectively, so as to realize the long-term preservation of key information. The structure of a single LSTM cell is illustrated in Fig. 13.

Principle of the LSTM is summarized as follows:

(1) The forget gate f_t is used to forget unimportant information from the previous cell state C_{t-1} , and it can be formulated as:

$$f_t = \text{sigmoid}(W_f[h_{t-1}, X_t] + b_f) \quad (14).$$

(2) The update of the cell state saves new information in cells, which mainly depends on the input gate i_t and $\tilde{C}_t$, where $\tilde{C}_t$ is a state value that determines how much input data can be retained. The input gate and state value are formulated as:

$$i_t = \text{sigmoid}(W_i[h_{t-1}, X_t] + b_i) \quad (15).$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, X_t] + b_C) \quad (16).$$

(3) After the first two gates, the network can decide which information to forget and which to save, and it can be written as:

$$C_t = f_t \times C_{t-1} + i_t \times \tilde{C}_t \quad (17).$$

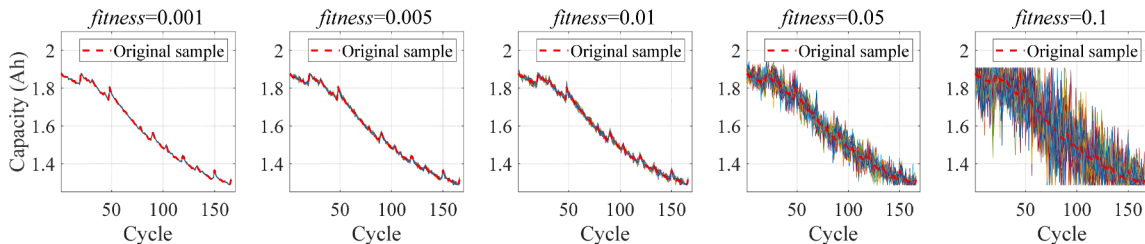
(4) Finally, the output of the estimation result (h_t) is decided by the output gate O_t as follows:

$$O_t = \text{sigmoid}(W_O[h_{t-1}, X_t] + b_O) \quad (18).$$

$$h_t = O_t \times \tanh(C_t) \quad (19).$$

Table 3 lists the detailed parameters of the LSTM neural network.

Outputs of the LSTM network are fed into a BP network to obtain the

Fig. 10. Virtual samples generated by different $fitness$.

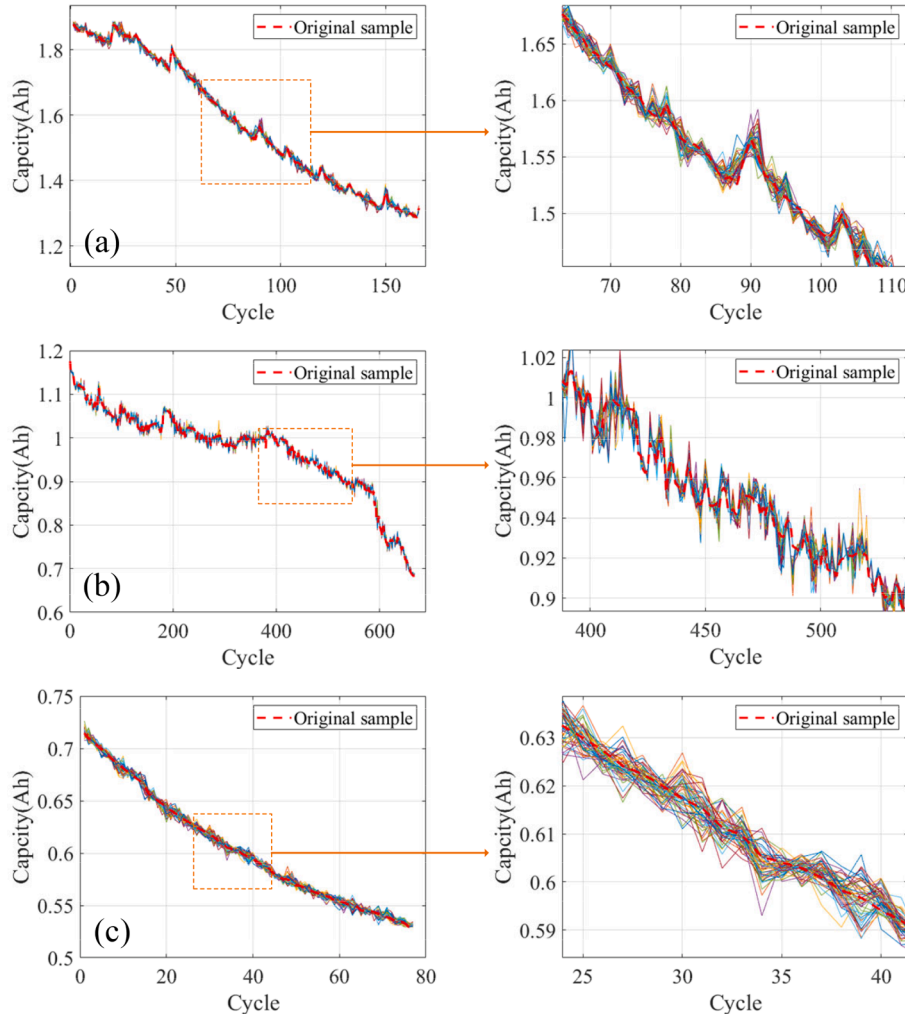


Fig. 11. Original and generated 50 virtual capacity curves: (a) NASA; (b) CALCE; (c) Oxford.

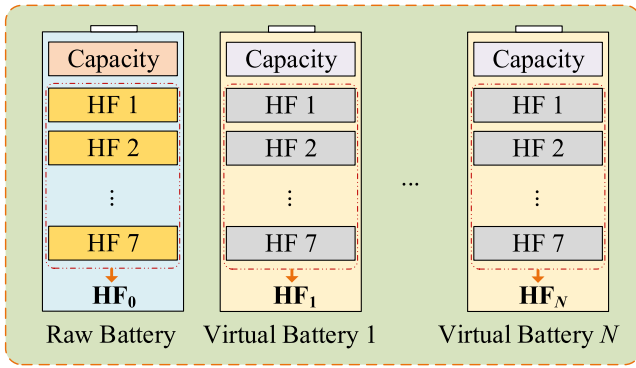


Fig. 12. Generated N groups of virtual samples.

estimated capacity. Input of the neural network is explained in Fig. 14, where HF_n and $capacity_n$ represents the seven HFs and capacity of the n -th cycle.

4. Results and discussion

To evaluate the feasibility and accuracy of the proposed method comprehensively, three indexes including root mean square error (RMSE), mean absolute percentage error (MAPE) and R^2 are utilized, and they can be formulated as follows:

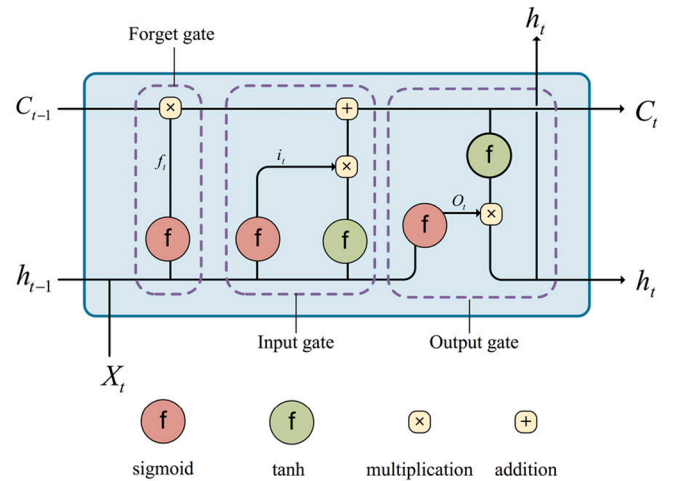


Fig. 13. The structure of a LSTM cell.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (q_i - \hat{q}_i)^2} \times 100\% (20).$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{q_i - \hat{q}_i}{q_i} \right| \times 100\% (21).$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (q_i - \hat{q}_i)^2}{\sum_{i=1}^N (q_i - \bar{q})^2} (22).$$

Table 3
Detailed parameters of the LSTM neural network.

Parameters	Value
Learning rate	0.01
Epochs	1000
Hidden size	128
LSTM layers	2
Time step	5
Batch size	64

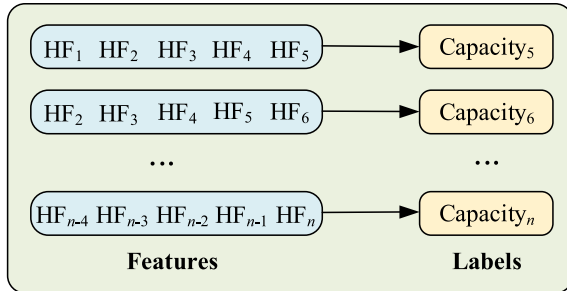


Fig. 14. The input data of neural network.

where q_i is the real capacity, $\hat{q}_i$ is the estimated capacity, $\bar{q}$ is the average value of the real capacity, and N is the total cycles.

4.1. Verification of voltage section optimization

To determine the optimal voltage section, the sum of correlation coefficient between each HF and capacity as shown in Eq. (2) is defined to be the objective, meanwhile constraints in Eq. (3) are added on both SOC and voltage range in consideration of the free charging behaviors of users in practical applications. Optimal voltage sections obtained by the QPSO algorithm are summarized in Table 4, where the “score” stands for the value of F in Eq. (2), and “time proportion” denotes the ratio of the sectional charging time to full charging time.

From Table 4, it can be seen that the charging time corresponding to the optimal voltage section is reduced obviously in comparison to the full charging time, and the time proportion is about 15 %–50 % only. Such a small-time proportion is benefit for reducing the data and time required for capacity estimation. Additionally, both the SOC and voltage range meet the constraints in Eq. (3). Nevertheless, the optimal sections

Table 4
Optimal voltage sections and details for each single battery.

Dataset	Battery	Optimal voltage section	Score	Start SOC	End SOC	Time proportion
NASA	B5	3.94 V–4.10 V	6.22	17.9 %	55.3 %	15.2 %
	B6	3.95 V–4.13 V	6.28	17.2 %	57.0 %	16.8 %
	B7	3.93 V–4.10 V	6.36	15.8 %	55.1 %	16.1 %
CALCE	CS35	3.90 V–4.09 V	6.23	23.2 %	68.2 %	35.1 %
	CS36	3.90 V–4.12 V	6.31	23.4 %	72.1 %	39.0 %
	CS37	3.91 V–4.13 V	6.33	24.6 %	72.9 %	37.3 %
Oxford	C1	3.74 V–3.94 V	6.53	21.0 %	67.5 %	46.5 %
	C3	3.74 V–3.95 V	6.50	21.5 %	69.9 %	48.4 %
	C8	3.74 V–3.94 V	6.54	21.8 %	68.0 %	46.3 %

of different cells for a same battery type are slightly different. Considering that it is impossible to set an optimal voltage section for each single battery individually in practical applications, the optimal voltage section needs to be unified for a same battery type. In this paper, the approximate intersection of the optimal section of each single battery is taken as the optimal section of the corresponding battery type, and the results are detailed in Table 5.

By comparing Table 5 to Table 4, it is clear that the score decreases slightly after unifying the optimal voltage section. Amongst, B7 of NASA is subjected to the biggest drop of 4.3 %, which decreases from 6.36 to 6.0. Overall, the decrease in score is slight, and its impact on later capacity estimation can be ignored.

In addition, due to the different charging modes used in the battery aging test, the ratio of the sectional charging time to the full charging time and SOC range of different battery types are inconsistent. For NASA dataset, there is a long constant voltage charging stage after the constant current charging stage. Therefore, the start SOC and end SOC corresponding to the optimal voltage section are small, and the sectional charging time is relatively short, with an average of 15.4 %. For CALCE dataset, the constant voltage charging stage is shorter than that of NASA dataset. Therefore, the start SOC and end SOC increase significantly compared with NASA dataset. The proportion of sectional charging time also increases significantly to an average of 37.6 %. For Oxford dataset, there is only a constant current charging process. Consequently, among the three battery types, it has the biggest ratio of the sectional charging time to full charging time, which is about 47 % averagely.

To demonstrate the superiority of the optimal voltage section, a non-optimal voltage section is adopted for comparison. The non-optimal voltage section is selected near the optimal voltage section, and its length is equal to the optimal voltage section. Details about the non-optimal voltage section are listed in Table 6.

The RMSE, MAPE and R^2 of capacity estimation with the full voltage section, the optimal voltage section and the non-optimal voltage section on the three battery datasets are compared in Table 7. It is evident from Table 7 that the full voltage curve achieves the highest accuracy of capacity estimation because it adopts the most amount of data. In contrast, due to the reduction of the used data, the sectional voltage curves lower the estimation accuracy. However, compared with the non-optimal voltage section, the optimal voltage section can improve the accuracy dramatically, even though they have the same voltage section length and approximate time proportion. For instance, the RMSE for NASA B6, CALCE CS36, and Oxford C3 batteries are improved from 3.48 % to 1.98 %, from 3.79 % to 2.99 %, and from 2.27 % to 0.70 %, respectively. In addition, the developed LSTM-BP neural network performs differently on the three datasets regardless of full or sectional charging curve. This is because the three datasets have highly different trends of capacity degradation as shown in Fig. 1(a), 2(a) and 3(a).

4.2. Verification of virtual sample generation

To further improve the generalization and accuracy of the data-

Table 5
Unified optimal voltage sections and details for each battery type.

Dataset	Unified optimal voltage section	Battery	Score	Start SOC	End SOC	Time proportion
NASA	3.94 V–4.10 V	B5	6.22	17.9 %	55.3 %	15.2 %
		B6	6.19	15.1 %	52.1 %	15.7 %
		B7	6.09	18.1 %	55.1 %	15.2 %
CALCE	3.91 V–4.13 V	CS35	6.10	26.4 %	74.3 %	38.1 %
		CS36	6.23	26.6 %	73.9 %	37.5 %
		CS37	6.33	24.6 %	72.9 %	37.3 %
Oxford	3.74 V–3.94 V	C1	6.53	21.0 %	67.5 %	46.5 %
		C3	6.48	21.5 %	69.6 %	48.1 %
		C8	6.54	21.8 %	68.0 %	46.3 %

Table 6

Non-optimal voltage section and details for each battery type.

Data set	Non-optimal voltage section	Battery	Score	Start SOC	End SOC	Time proportion
NASA	3.89 V-4.05 V	B5	2.98	9.8 %	46.2 %	14.9 %
		B6	2.87	8.1 %	42.4 %	14.6 %
		B7	3.16	10.0 %	46.1 %	14.9 %
CALCE	3.95 V-4.17 V	CS35	4.35	38.4 %	79.9 %	32.9 %
		CS36	4.68	38.3 %	79.5 %	32.5 %
		CS37	4.52	36.3 %	78.5 %	33.1 %
Oxford	3.84 V-4.04 V	C1	3.18	44.6 %	82.1 %	37.5 %
		C3	2.69	45.2 %	82.2 %	37.1 %
		C8	2.73	45.4 %	82.3 %	36.9 %

driven model, this paper employs the MD-MTD-QPSO algorithm as introduced in Section 3.3 to generate 50 sets of virtual battery samples for training the model. This section compares the estimated results with and without virtual samples in detail.

For NASA dataset, the sectional data obtained from the optimal voltage section of battery B5 are used to extract the HF. Based on the originally extracted HF, 50 sets of virtual battery samples are generated and used to train the model together. To verify the effectiveness of the virtual samples generation for improving the estimation accuracy, the optimal voltage section data of batteries B6 and B7 are used as test datasets. The comparison results of battery B6 with and without virtual samples are shown in Fig. 15. It can be seen that by introducing the generated virtual samples, the average error of battery B6 can be reduced from 0.024 Ah to 0.011 Ah, with an average decrease of 54 %. The estimation results of NASA B7 with and without virtual samples are shown in Fig. 16. It is clear that compared with the result without virtual samples, the average error of battery B7 with virtual samples is lowered by 41 %, from 0.017 Ah to 0.010 Ah.

For CALCE dataset, the sectional data corresponding to the optimal voltage section of battery CS35 are used to extract the HF. Then 50 sets of virtual battery samples are generated and utilized to train the model

along with the original HF extracted from battery CS35. CS36 and CS37 battery datasets are adopted to test the model. Fig. 17 shows the capacity estimation results of battery CS36 with and without virtual samples. With the generated virtual samples, the average error of battery CS36 is lowered from 0.017 Ah to 0.008 Ah, i.e., an average decrease of 53 %. Fig. 18 shows the capacity estimation results of battery CS37 with and without virtual samples. It can be seen that the average error of battery CS37 without virtual samples is 0.023 Ah, while that with virtual samples is 0.011 Ah. That is, the average error is decreased by 52 %.

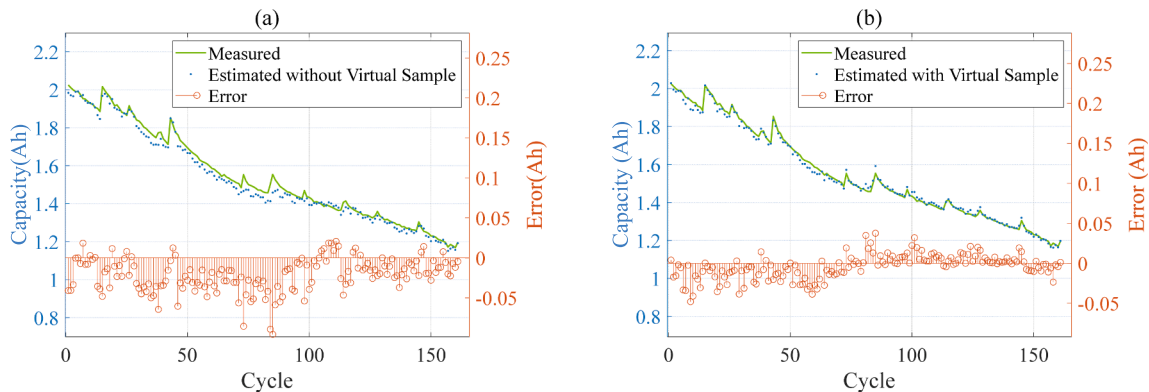
For Oxford dataset, HF are firstly extracted from the sectional data corresponding to the optimal voltage section of battery C1. Then the extracted HF are employed to generate 50 sets of virtual battery samples, which are applied to train the neural network model along with the HF originally extracted from battery C1. To compare the model accuracy with and without virtual samples, the optimal voltage section data of batteries C3 and C8 are used for test datasets. Capacity estimation results of battery C3 with and without virtual samples are plotted in Fig. 19 and capacity estimation results of battery C8 with and without virtual samples are plotted in Fig. 20. It is indicated that after introducing the virtual samples, the average error of batteries C3 and C8 can be reduced from 0.0035 Ah to 0.0016 Ah, and from 0.0039 Ah to 0.0027 Ah, respectively. In other words, for these two batteries, the average error decrease by about 54 % and 31 %, respectively.

RMSE, MAPE and R^2 for the three battery datasets with and without generated virtual samples are summarized in Table 8 to compare the capacity estimation results more clearly. It is evident that both the RMSE and MAPE are reduced significantly, while the R^2 is increased after adding the virtual samples regardless of the battery type or cell. Particularly, the average RMSE decreases by 48 %, the average MAPE decreases by 51 %, and the average R^2 increases by 0.01. Amongst, the CALCE dataset achieves the highest accuracy improvement with an average RMSE of 51 % and an average MAPE of 54 %. Additionally, the estimation accuracies with virtual samples are higher than that using the full charging curve in Table 7. For instance, the RMSEs for batteries NASA B6, CALCE CS36, and Oxford C3 are reduced from 1.56 % to 1.05

Table 7

Comparison results of capacity estimation with different voltage sections.

Battery	Full voltage curve			Non-optimal voltage section			Optimal voltage section		
	RMSE	MAPE	R^2	RMSE	MAPE	R^2	RMSE	MAPE	R^2
NASA B6	1.56 %	1.21 %	0.988	3.48 %	2.89 %	0.967	1.98 %	1.59 %	0.984
NASA B7	0.95 %	0.86 %	0.991	3.59 %	2.47 %	0.973	1.26 %	1.09 %	0.983
CALCE CS36	2.12 %	1.98 %	0.997	3.79 %	3.82 %	0.955	2.99 %	2.83 %	0.991
CALCE CS37	2.22 %	2.01 %	0.988	4.61 %	4.39 %	0.940	3.68 %	3.05 %	0.962
Oxford C3	0.52 %	0.45 %	0.996	2.27 %	2.15 %	0.976	0.70 %	0.61 %	0.993
Oxford C8	0.63 %	0.52 %	0.993	3.48 %	3.57 %	0.946	0.72 %	0.65 %	0.992

**Fig. 15.** Capacity estimation results of NASA B6: (a) without virtual samples; (b) with virtual samples.

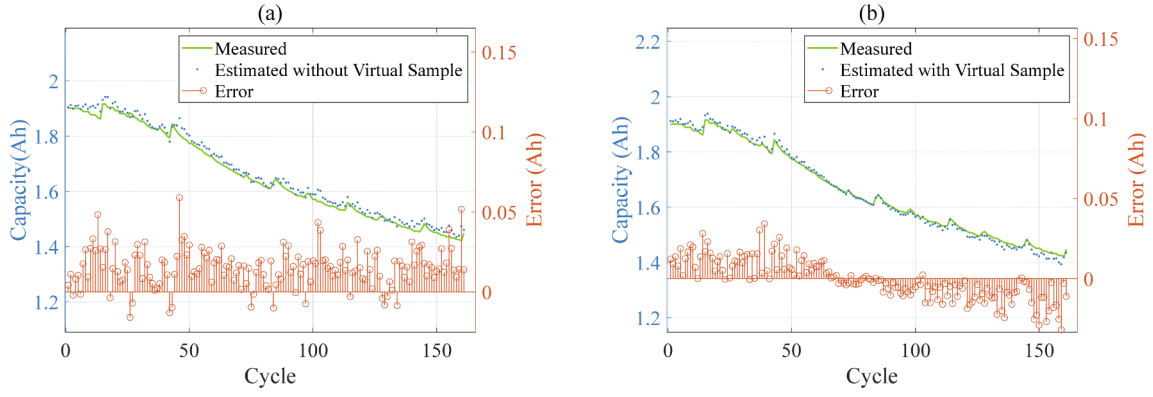


Fig. 16. Capacity estimation results of NASA B7: (a) without virtual samples; (b) with virtual samples.

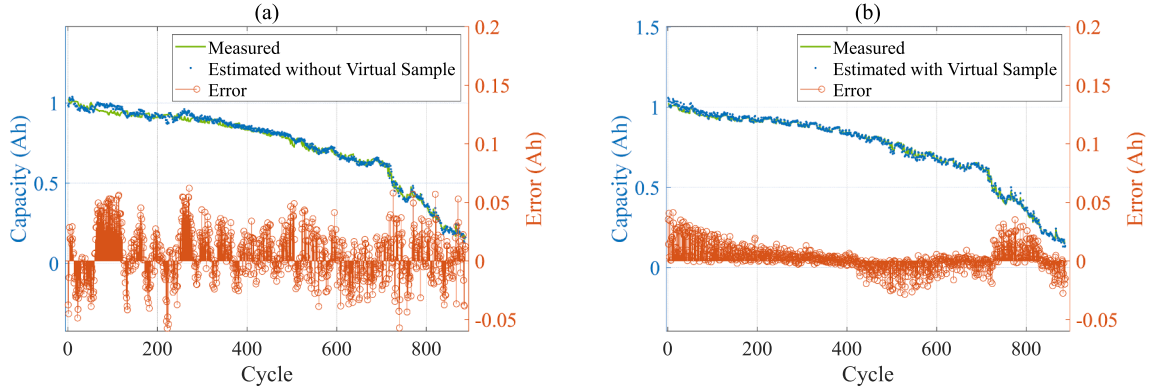


Fig. 17. Capacity estimation results of CALCE CS36: (a) without virtual samples; (b) with virtual samples.

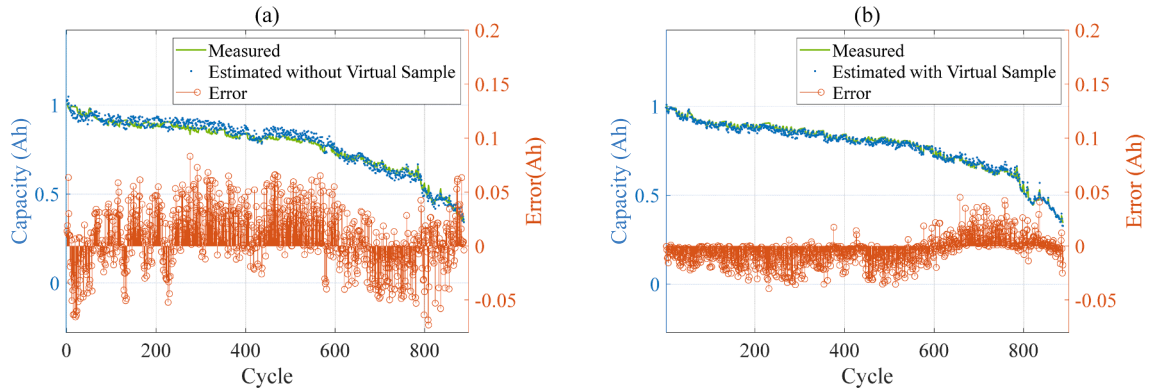


Fig. 18. Capacity estimation results of CALCE CS37: (a) without virtual samples; (b) with virtual samples.

%, from 2.12 % to 1.48 %, and from 0.52 % to 0.29 %, respectively. Therefore, it can be concluded that the proposed method can improve the model accuracy and reduce the amount of required data for capacity estimation simultaneously.

5. Conclusions

In consideration of the incomplete charging process and lacking sufficient datasets for training a data-driven battery capacity model, a battery capacity estimation method based on charging voltage section and virtual sample generation is proposed in this paper. Health factors are constructed from both model domain and data domain to capture the law of capacity degradation more accurately and robustly. To determine an optimized charging voltage section, a nonlinear programming model

that is to maximize the sum of the absolute Pearson coefficient between each health factor and the capacity is established and a quantum particle swarm optimization is employed for finding the global optimization resolution. A multi-distribution global trend diffusion algorithm is introduced to generate virtual samples based on existing battery test data, thus augmenting the datasets for model training. A hybrid long short-term memory and back-propagation neural network is constructed to learn the nonlinear relationship between the health factors and the capacity. Three public datasets are employed to verify the generalization and accuracy of the proposed method. Experimental results indicate that the proposed method can reduce the amount of required data for battery capacity estimation, and the average charging time of the optimal voltage section only accounts for 33.3 % of the full charging time. Compared with a non-optimal voltage section, the optimal voltage

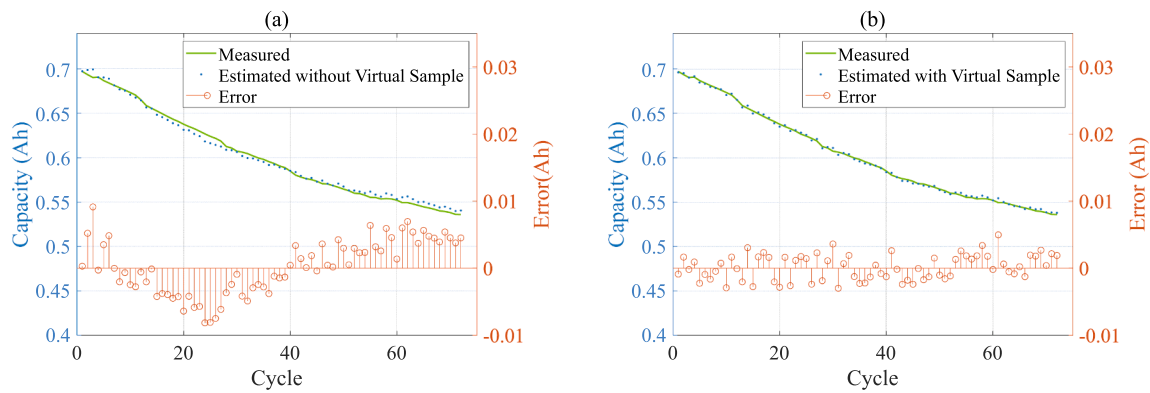


Fig. 19. Capacity estimation results of Oxford C3: (a) without virtual samples; (b) with virtual samples.

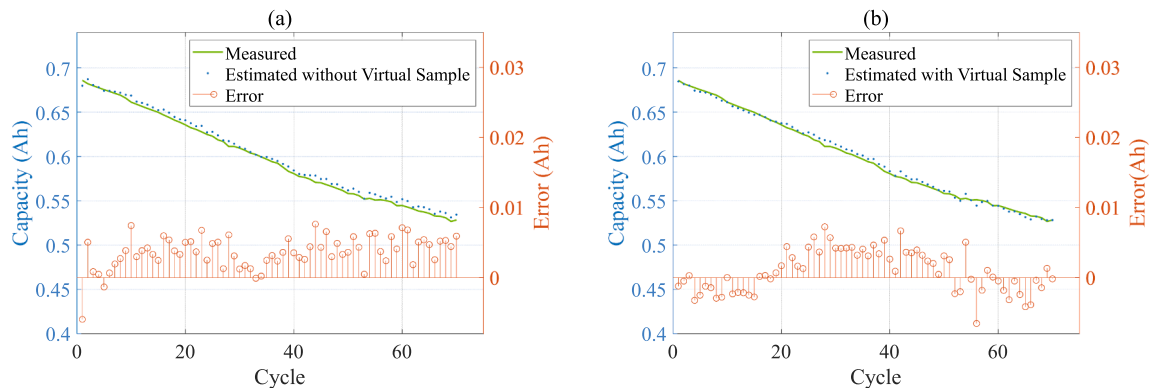


Fig. 20. Capacity estimation results of Oxford C8: (a) without virtual samples; (b) with virtual samples.

Table 8

Comparison of capacity estimation results with and without virtual samples.

Battery	Without virtual samples			With virtual samples		
	RMSE	MAPE	R ²	RMSE	MAPE	R ²
NASA B6	1.98 %	1.59 %	0.984	1.05 %	0.77 %	0.996
NASA B7	1.26 %	1.09 %	0.983	0.74 %	0.59 %	0.994
CALCE CS36	2.99 %	2.83 %	0.991	1.48 %	1.31 %	0.998
CALCE CS37	3.68 %	3.05 %	0.962	1.76 %	1.37 %	0.991
Oxford C3	0.70 %	0.61 %	0.993	0.29 %	0.25 %	0.999
Oxford C8	0.72 %	0.65 %	0.992	0.53 %	0.45 %	0.996

section can improve the estimation accuracy dramatically, even though they have the same voltage section length and approximate time proportion. In addition, the capacity estimation errors with virtual samples are even smaller than that using the full charging voltage curve. For instance, the root mean square errors for batteries NASA B6, CALCE CS36, and Oxford C3 are reduced from 1.56 % to 1.05 %, from 2.12 % to 1.48 %, and from 0.52 % to 0.29 %, respectively. It can be concluded that the proposed method can improve the battery capacity estimation accuracy and reduce the amount of required data simultaneously.

The method proposed in this paper has a good estimation accuracy on the same type of batteries with the same charging process. However, the limitation of this method is that it cannot estimate the capacity for different types of batteries in one model, which will limit the application scope of this model. Moreover, battery charging processes have a strong randomness and uncertainty, the method will be invalid when the optimal voltage section cannot be covered by user's charging voltage section. In future work, we will employ some methods to solve the above problems.

CRediT authorship contribution statement

Yong Tian: Conceptualization, Methodology, Software, Validation, Writing – original draft, Project administration, Funding acquisition. **Qianyu Dong:** Software, Validation, Data curation, Writing – original draft. **Jindong Tian:** Conceptualization, Supervision, Writing – review & editing. **Xiaoyu Li:** Conceptualization, Writing – review & editing, Funding acquisition. **Guang Li:** Conceptualization, Writing – review & editing. **Kamyar Mehran:** Formal analysis, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

This work was supported by the open research fund from Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ) (No. GML-KF-22-19), the National Natural Science Foundation of China (No. 52177219) and the Natural Science Foundation of Guangdong Province (2021A1515010525). We also thank the NASA Ames Research Center, the Center for Advanced Life Cycle Engineering (CALCE) of the University of Maryland, and the battery Intelligence Laboratory of Oxford University for providing the battery datasets. which could be accessed on the website: <https://data.nasa.gov/dataset/Li-ion-Battery-Aging-Datasets/uj5r-zjdb>, <https://calce.umd.edu/data#CS2> and <https://>

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